Tahsin Touheed Ivan

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Education

University of Dhaka — Dhaka, Bangladesh
B.Sc. in Electrical and Electronic Engineering (4th year) — *Sept 2022 – Jan 2027 (expected)*

CGPA: 3.73 / 4.00

Relevant Coursework: Quantum Mechanics, Materials Science, Solid State Physics, Electronic Devices, Device Fabrication Techniques, Optoelectronics & Photonics, Modern Physics, Chemistry, Linear Algebra & Numerical Analysis, Probability & Statistics.

Notre Dame College — Dhaka, Bangladesh
Higher Secondary Certificate (Science) — *Aug 2019 – July 2021*

GPA: 5.00 / 5.00

Ideal School & College — Dhaka, Bangladesh
Secondary School Certificate (Science) — *Jan 2009 – July 2019*

GPA: 5.00 / 5.00

Research

Materials Science Research Laboratory (MSRL DU) — Jan 2026 – Present
Undergraduate Research Assistant* | *Supervisor: Prof. Dr. Imtiaz Ahmed

Research Interests: AI/ML-assisted materials discovery, Computational Materials Engineering, Energy storage devices, Nanomaterials synthesis & characterization, Density Functional Theory; with broader curiosity in Micro/Nano-engineered devices.

Current Work — Machine-Learning & DFT-Guided Electrode Design for Oxide-based Supercapacitors: From Data Mining to Experimental Validation

- Building a computation-guided closed-loop pipeline for perovskite-oxide (ABO_3) based supercapacitor electrodes. Stage 1 was targeted towards curating a literature-mined descriptor dataset, that consisted 889 unique data (across 86 different published experimental works), while the feature space spanning across chemistry, structure, processing and testing-condition. Then built a classifier model ensemble that rank compositions by propensity of top-tier specific capacitance (specifically, top 10% of the data space); utilised SHAP/PDP explainability to identify the most dominant drivers, yielding a ranked shortlist of undoped and doped candidates.
- A stability screening model narrows the shortlist before first-principles validation, concentrating compute on the most viable candidates; surviving compositions enter a DFT workflow establishing the true ground state and confirming mechanical (elastic constants, Born criteria) and dynamical (phonon) stability, followed by charge-storage descriptors — quantum capacitance from the density of states, ion adsorption energy with Bader charge transfer, and volume expansion on ion insertion.
- Stage 2 targets experimental validation: screened candidates synthesized toward sub-200 nm particles for near-surface Faradaic utilisation, electrodes fabricated by spray pyrolysis or laser-induced imprinting techniques, structurally characterized (XRD, BET, Raman, XPS) and electrochemically tested (CV, GCD, EIS), with measurements fed back to refine the model; closing the loop.

Leadership

IEEE SIGHT Student Branch, DU — Feb 2025 – Present

Chair — *April 2026 – Present*

- Led the team that secured the IEEE HTB Tech4Good 2026 grant (USD 4,739) for *SmartDrip*, an IoT-enabled drip irrigation system for urban rooftop plantation; currently leading project execution end-to-end.

Program Coordinator — *Feb 2025 – March 2026*

- Coordinated and organized the *Machine Learning Fundamentals* workshop for undergraduate students — managed logistics, content planning, and participant engagement.

- Led a hands-on session in the *IEEE Day Outreach Programme on Microcontrollers and Their Utilization* — jointly organized by IEEE SB University of Dhaka, IEEE EDS SBC DU, and IEEE WIE AG SB DU at Nawab Faizunnesa Govt Girls High School, Cumilla — for 50+ high school students.

Optics and Photonics Club — Jan 2024 – April 2025

Vice President — *Jan 2024 – April 2025*

- Delivered talks and project showcases (LFR, Holography, etc.) at the outreach program *Envisioning Harmony Between Human and Nature*, St. Joseph Higher Secondary School, Dhaka — in collaboration with the Optica Bangladesh Student Chapter, University of Dhaka, and Scintilla Science Club.

Projects

SmartDrip: IoT-Enabled Drip Irrigation System for Urban Rooftop Plantation | *Embedded Systems* — 2026 – Ongoing

Awarded: *IEEE HTB Tech4Good 2026 Grant (USD 4,739)*

- Leading an IEEE SIGHT SB DU team developing an IoT-enabled drip irrigation system aimed at making urban rooftop plantation water-efficient and low-maintenance for Old Dhaka residential area — addressing freshwater stress and food-resilience challenges in dense urban areas.
- Proposal selected for funding under the IEEE Humanitarian Technologies Board's Tech4Good 2026 call; currently directing system design, sensor–actuator integration, and field-deployment planning.

LC-Based Capacitive MEMS Pressure Microsensor | *Python, scikit-learn* — 2026

- Built an end-to-end Python simulation of a passive, wirelessly-interrogated capacitive MEMS pressure sensor — modeling the full signal chain from membrane deflection to variable-gap capacitance, LC resonance, and inductive readout.
- Trained a neural-network surrogate on 171 Latin-Hypercube-sampled geometries and applied inverse design to optimize sensor geometry for target performance (∼600 times faster than the full simulator, every candidate verified against it) — the same screen-and-validate strategy being applied at materials scale in my current ML/DFT based research work at MSRL.

Project: Drone (Funded by FabLab DU) | *Python, Arduino, ESP32* — 2025

Supervised by: *Dr. Md. Shafiul Alam*

- Co-developed FabLab DU's first departmental drone for ∼USD 66 — integrated propulsion, IMU, and RC link on an ESP32 flight controller, hand-tuned cascaded PID control across all three axes, and resolved ESC desync and vibration instability to achieve 2.8:1 thrust-to-weight and had achieved ∼9 min flight time on its first field test.

ESP32 Retro Gaming Console | *ESP32* — 2026

- Built a handheld console prototype (ESP32, OLED, analog joystick) running three original arcade games with a scrollable menu and persistent high scores — real-time firmware built on millis()-based finite-state machines, ADC filtering with hysteresis, and non-volatile storage across deep-sleep cycles.

Pocket Reader — Handheld E-Book Reader | *ESP32* — 2026

- Built a self-contained e-book reader prototype serving a mobile-friendly web portal directly from the microcontroller for uploads and Wi-Fi management — no cloud, no external server; on-device REST server with mDNS, LittleFS storage, and deep-sleep power cycling.

Laser Harp | *Arduino* — 2023

- Academic project from sophomore year — played a two-octave span of musical notes using a laser-and-LED array.

Competencies

Programming: Python (scikit-learn, pandas, NumPy, SciPy, SHAP, Matplotlib, Keras/TensorFlow), C/C++, x86 Assembly, Verilog (RTL design), HTML, LaTeX.

Software & Tools: VASP, Quantum Espresso, MATLAB, PSPICE, Proteus, AutoCAD, EasyEDA.

Languages: English (IELTS 7.0; test taken on 2023), Bangla (native).

Soft Skills: Public Speaking, Team Leadership and Mentoring.

References

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Additional references available upon request.